

Hotel Network Design – Technical Documentation

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Devices & Roles:

Device Name	Device Type	Role
R-FLOOR-1	Router 2811	Inter-VLAN routing, DHCP, OSPF (Floor 1)
R-FLOOR-2	Router 2811	Inter-VLAN routing, DHCP, OSPF (Floor 2)
R-FLOOR-3	Router 2811	Inter-VLAN routing, DHCP, OSPF (Floor 3)
S-FLOOR-1	Switch 2960	VLAN segmentation & trunking (Floor 1)
S-FLOOR-2	Switch 2960	VLAN segmentation & trunking (Floor 2)
S-FLOOR-3	Switch 2960	VLAN segmentation & trunking (Floor 3)
Wireless APs	Access Points	Wireless connectivity per floor
PCs, Laptops, Printers, Mobiles	End Devices	Connected to respective department VLANs

Floor-wise VLAN & IP Addressing Plan

Floor	Department	VLAN ID	Network	Default Gateway
1	Reception	80	192.168.8.0/24	192.168.8.1
1	Store	70	192.168.7.0/24	192.168.7.1
1	Logistics	60	192.168.6.0/24	192.168.6.1
2	Finance	50	192.168.5.0/24	192.168.5.1
2	HR	40	192.168.4.0/24	192.168.4.1
2	Sales	30	192.168.3.0/24	192.168.3.1
3	Admin	20	192.168.2.0/24	192.168.2.1
3	IT	10	192.168.1.0/24	192.168.1.1

Inter-VLAN Routing

R-FLOOR-1

Subinterface	VLAN ID	IP Address
Fa0/0.10	10	192.168.10.1
Fa0/0.60	60	192.168.6.1
Fa0/0.70	70	192.168.7.1
Fa0/0.80	80	192.168.8.1

R-FLOOR-2

Subinterface	VLAN ID	IP Address
Fa0/0.10	10	192.168.20.1
Fa0/0.30	30	192.168.3.1
Fa0/0.40	40	192.168.4.1
Fa0/0.50	50	192.168.5.1

R-FLOOR-3

Subinterface	VLAN ID	IP Address
Fa0/0.10	10	192.168.1.1
Fa0/0.20	20	192.168.2.1

DHCP Pool Summary

Router	DHCP Pool	Subnet
R-FLOOR-1	ReceptionPool	192.168.8.0/24
R-FLOOR-1	StorePool	192.168.7.0/24
R-FLOOR-1	LogisticsPool	192.168.6.0/24
R-FLOOR-2	FinancePool	192.168.5.0/24
R-FLOOR-2	HRPool	192.168.4.0/24
R-FLOOR-2	SalesPool	192.168.3.0/24
R-FLOOR-3	AdminPool	192.168.2.0/24
R-FLOOR-3	ITPool	192.168.1.0/24

Switch Port Mapping

S-FLOOR-1

Switch Port	VLAN	Connected Device
Fa0/1	Trunk	Link to R-FLOOR-1
Fa0/2	80	Reception PC
Fa0/3	80	Reception Printer
Fa0/4	70	Store PC
Fa0/5	70	Store Printer
Fa0/6	60	Logistics PC
Fa0/7	60	Logistics Printer
Fa0/8	10	Wireless AP

S-FLOOR-2

Switch Port	VLAN	Connected Device
Fa0/1	Trunk	Link to R-FLOOR-2
Fa0/2	50	Finance PC
Fa0/3	50	Finance Printer
Fa0/4	40	HR PC
Fa0/5	40	HR Printer
Fa0/6	30	Sales PC
Fa0/7	30	Sales Printer
Fa0/8	10	Wireless AP

S-FLOOR-3

Switch Port	VLAN	Connected Device
Fa0/1	Trunk	Link to R-FLOOR-3
Fa0/2	20	Admin PC
Fa0/3	20	Admin Printer
Fa0/4	10	Test-PC (Port Security Enabled)
Fa0/5	10	IT PC
Fa0/6	10	Wireless AP

OSPF Inter-Router Networks

Link	Network
R-FLOOR-1 ↔ R-FLOOR-2	10.10.10.0/30
R-FLOOR-2 ↔ R-FLOOR-3	10.10.10.4/30
R-FLOOR-1 ↔ R-FLOOR-3	10.10.10.8/30

Port Security Configuration

Port	VLAN	Security Method	Violation Mode
Fa0/4	10	Sticky MAC	Shutdown

Wireless Network Configuration

Floor	VLAN	SSID	Security	Connected Devices
Floor 1	10	FLOOR1-WIFI	WPA2-PSK	Laptops, Mobiles
Floor 2	10	FLOOR2-WIFI	WPA2-PSK	Laptops, Mobiles
Floor 3	10	FLOOR3-WIFI	WPA2-PSK	Laptops, Mobiles